

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 2

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(210 - 12) / 210 \times 100$ (1) • = 94.3% (accept 94%) (1)	2
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Conserves mineral reserves / extends their lifetime (1) • Reduces the environmental impacts of mining (land take, habitat loss, spoil, dust, noise, turbid water) (1) • Reduces mineral processing pollution (eg red mud from bauxite refining) (1) • Reduces waste disposal impacts / landfill volume (1) • Less CO₂ / climate change (1)	3
01.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Identification of materials: slow, labour-intensive to determine composition where it matters (eg alloys, plastic types) (1) • Separation of mixed materials: alloys cannot easily be separated so recycled aluminium cannot be used where pure metal is needed; mixed wastes reduce quality (1) • Reduction in quality / contamination (eg paper mixed with food waste) (1) • Transport / collection: small dispersed quantities cannot use bulk transport so costs and energy per tonne are higher (may exceed savings) (1) • Lack of consumer cooperation in separating wastes; waste losses to litter and general waste; labour costs (1)	4
01.4	<i>1 mark.</i> • Designing products so that the materials can be re-used at the end of their lives - easy separation of components and identification of materials - extending the lifetime of mineral use (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 146-147; AQA spec 3.2.3.7.	
02.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
02.2	<i>1 mark.</i> • All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	